

Design and Simulation of Monostable Multivibrator Using IC 555

Objectives

- To study the operational behavior and internal architecture of the IC 555 timer configured as a Monostable Multivibrator.
- To design a monostable circuit using LTspice with specific component parameters ($R=4.7\text{ k}\Omega$, $C=2.2\text{ }\mu\text{F}$) and verify its theoretical timing characteristics against simulation outputs.
- To analyze how negative edge-triggering pulses activate the quasi-stable state and measure the resulting output pulse width.

Theory

A monostable multivibrator, often called a one-shot multivibrator, is a pulse-generating circuit with only one stable state (logic low). Upon application of an external negative-going trigger pulse to Pin 2 (Trigger), the internal circuitry transitions into a temporary quasi-stable state (logic high)

Principle of Operation

1. **Stable State:** Initially, the output (Pin 3) is low, turning the internal discharge transistor ON. This shorts out the timing capacitor to ground, keeping it fully discharged.
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2. **Triggering:** When a negative trigger pulse drops below $\frac{1}{3} V_{CC}$ at Pin 2, the lower internal comparator trips, changing the state of the RS flip-flop. The internal discharge transistor turns OFF, releasing the short across the external timing capacitor, and the output (Pin 3) switches to a high state ($\approx V_{CC}$).
3. **Timing and Recovery:** The timing capacitor begins charging toward V_{CC} exponentially through the resistor R_1 . When the capacitor voltage matches or exceeds the threshold level ($\frac{2}{3} V_{CC}$) at Pin 6 (Threshold), the upper comparator resets the RS flip-flop. The output abruptly

reverts to its stable low state, the internal transistor turns back ON, and the capacitor rapidly discharges to ground.

Mathematical Design Equations

The precise duration of the high output pulse (T) depends solely on the external RC timing network and is derived as follows:

$T=1.1\times R\times C$

Circuit Design & Component Values

Based on the implemented LTspice schematic, the circuit utilizes a V CC =5V DC power supply.

Component Label	Type / Description	Value	Purpose
U1	IC Timer	NE555	Core timing integrated circuit
R1	Timing Resistor	4.7 kΩ	Governs capacitor charging current
C2	Timing Capacitor	2.2 μF	Stores charge to define pulse duration
C1	Bypass Capacitor	0.01 μF	Stabilizes the control voltage terminal (Pin 5) against noise
R2	Output Load Resistor	10 kΩ	Models the standard output load line
V1	DC Power Source	5V	Main circuit supply voltage (VCC)
V2	Triqger Pulse Source	Configured Pulse	Simulates the input triqgering pulse string

Theoretical Pulse Width Calculation Using the designated schematic values:

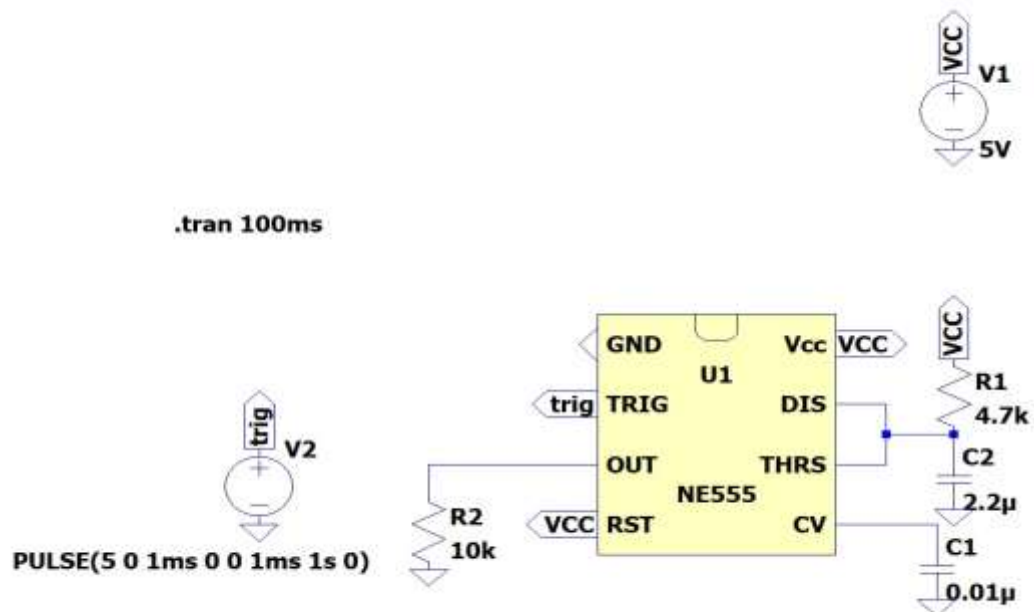
$R\ 1\ =4.7\ k\Omega=4700\ \Omega$

$C\ 2\ =2.2\ \mu F=2.2\times10^{-6}\ F$

$T=1.1\times4700\times(2.2\times10^{-6})\approx0.011374\ seconds\approx11.37\ ms$

LTspice Simulation Setup

- The Circuit

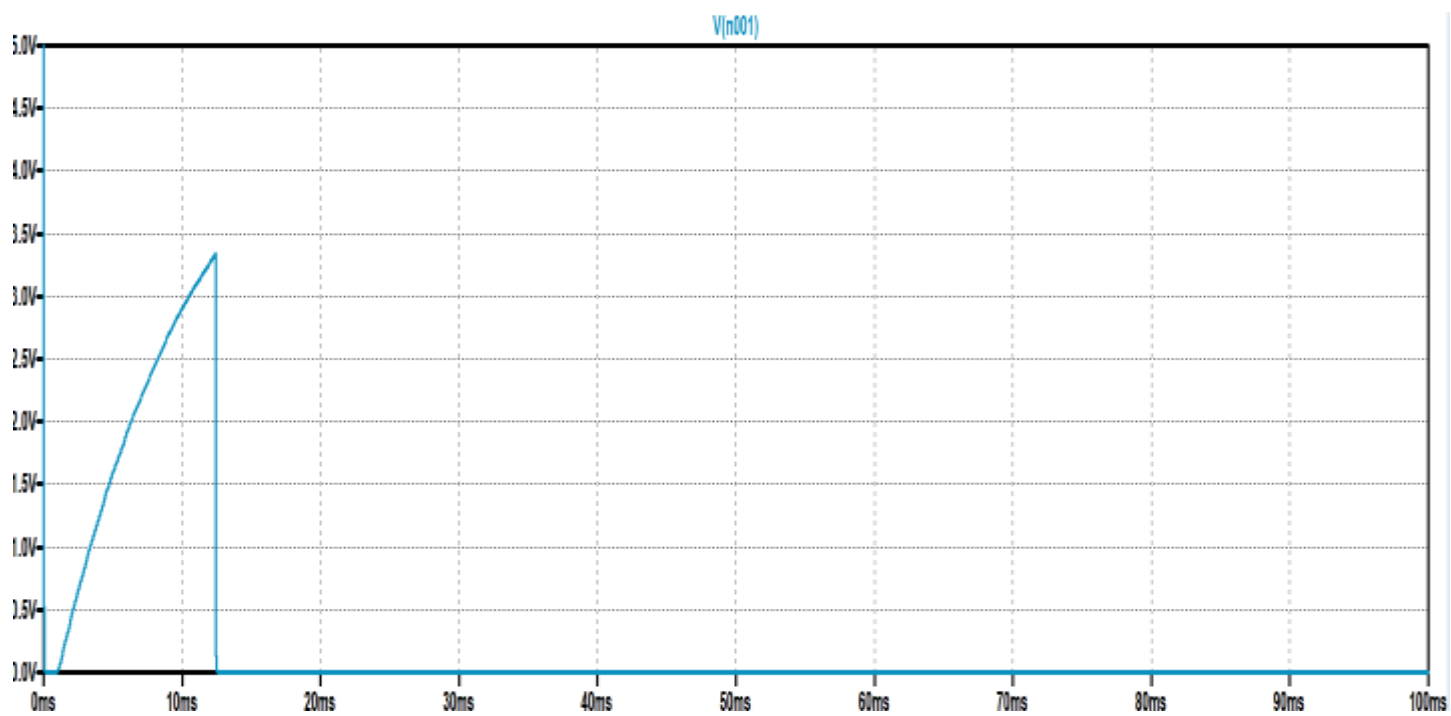
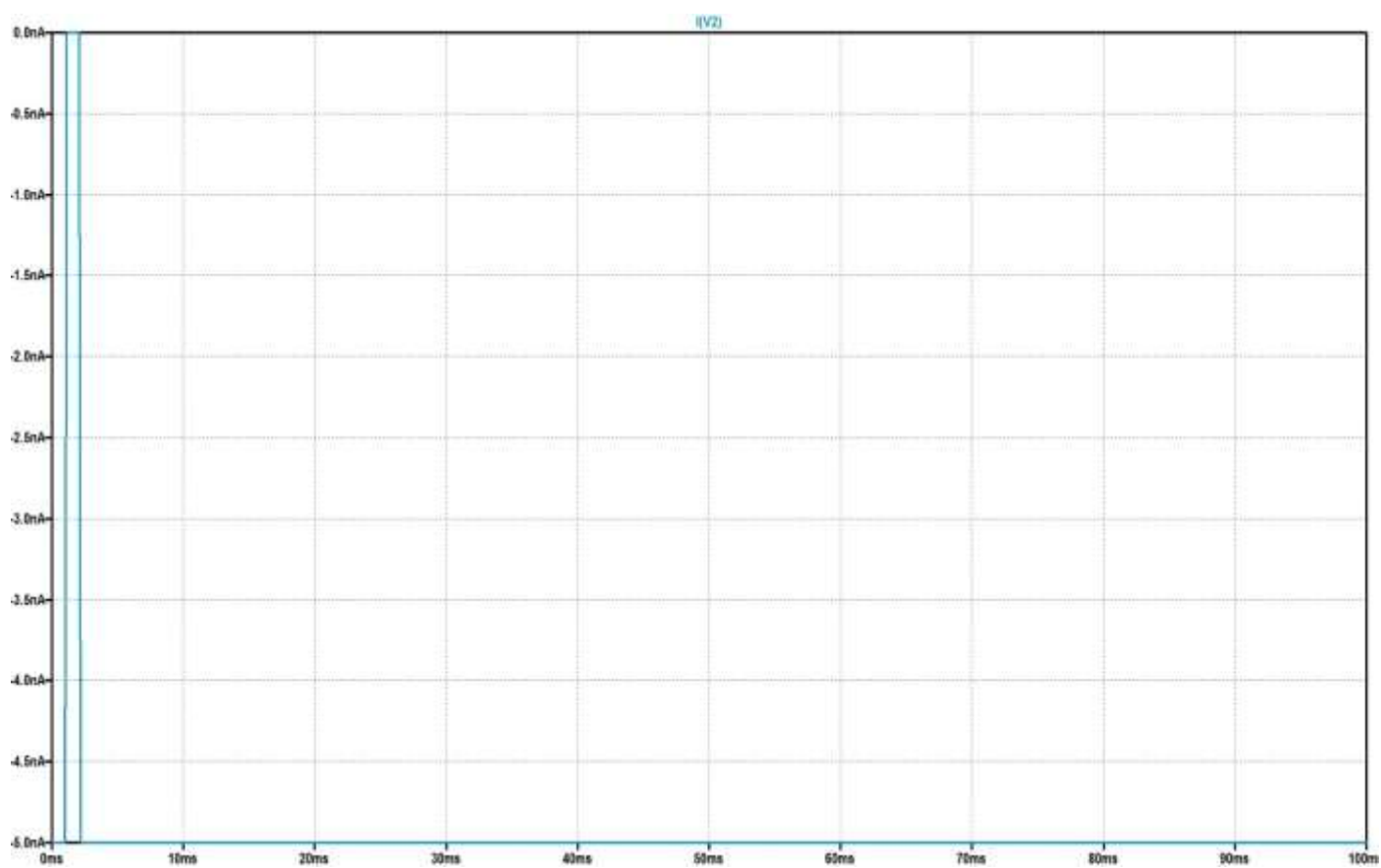


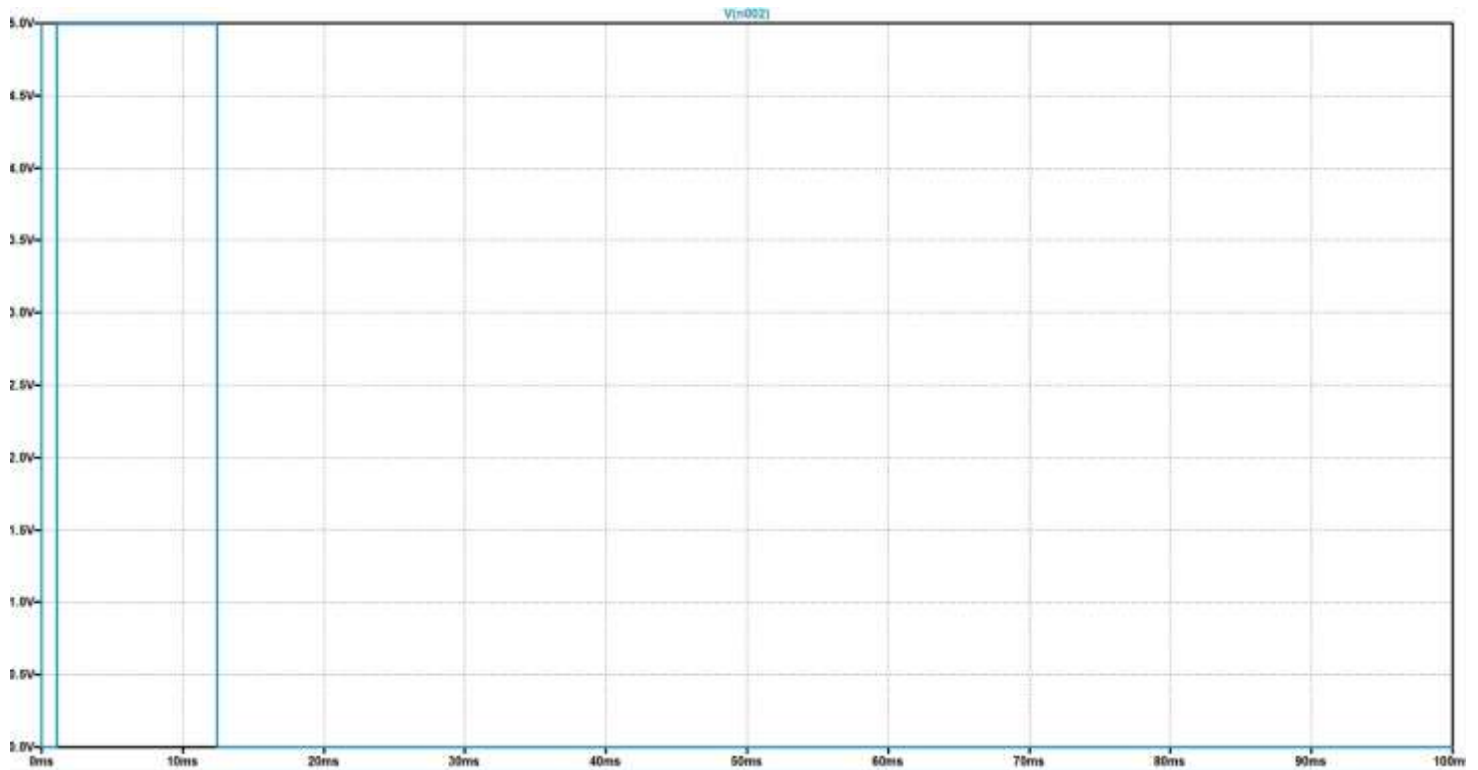
To successfully verify the hardware model, the following parameters were coded into the independent pulse generator V2 to issue a single crisp negative-edge trigger:

- Initial Voltage (V initial): 5V (Keeps the trigger inactive at V CC)
- On Voltage (V on): 0V (Drops down to ground to activate the lower comparator)
- Delay Time (T delay): 1 ms
- Pulse Width (Ton): 1 ms
- Period (T period): 1 s

Simulation Command: .tran 100ms (Captures transient behavioral curves through a window of 100 ms)

Results and Analysis





Waveform Parameter Verification By tracking the transient graphs plotted from LTspice, we extract the following measurements:

- Trigger Timing: The input signal V2 falls from 5V to 0V at $t=1$ ms.
- Capacitor Charging: At $t=1$ ms, the capacitor voltage V(n001) immediately begins an exponential rise from 0V toward its targeted threshold value.
- **Threshold Turn-off: The capacitor successfully charges until it climbs to $2/3$ V CC**
- $2/3 * (5V) \approx 3.33V$. The graph confirms a sharp drop back to 0V right when it hits this 3.33V benchmark.

Output Pulse Width Measurement: The output voltage V(n002) jumps to high (5V) at exactly $t=1$ ms and drops back to low (0V) at approximately $t=12.37$ ms.

$$\Delta T_{\text{observed}} = 12.37 \text{ ms} - 1.00 \text{ ms} = 11.37 \text{ ms}$$

Parameter Comparison Table

Parameter	Theoretical Value	Simulated Value	Error (%)
Output Pulse Duration (T)	11.37 ms	11.37 ms	0%
Threshold Trip Voltage	3.33V	3.33V	0%

Inference

The simulation strictly aligns with theoretical monostable behavior. The output pulse width matches the mathematical estimation exactly (11.37 ms), showing a 0% margin of error. This proves that our LTspice timing model operates properly under ideal software conditions.

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The scope sweeps perfectly illustrate that the output remains independent of any trigger behaviors after activation, remaining high until the tracking network hits its

3.2 V_{CC} threshold ceiling.